

期中复习

1. $\varepsilon-N$

必须放缩成 $M \cdot n$, 分母如果出现 n^2 , 放缩成 $\frac{1}{n^2}$; $\dots \frac{1}{n^k} \rightarrow \frac{1}{n^k}$; $\frac{1}{3^n}$ 二项式展开: $q^n (|q| < 1)$ 将 q 换为 $\frac{1}{n}$ 二项式展开, 分母比分多一次方.

2. $\varepsilon-\delta$

限定 $|x-x_0| < \delta_0$, 放缩为 $M|x-x_0|^\alpha (\alpha > 0) < \varepsilon$, 但去绝对值时要小心, $\delta = \min\{\delta_0, \text{带}\varepsilon\text{的}\}$.

eg. 证 $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1$

3. 夹逼定理 单调有界 (递推表达式)

有单调性找界, 无单调性找猜极限解出, 再用 $\varepsilon-N$ 证明其收敛. 放缩成 $a^{n-1}|x_n-A|, |x_n-A|$

证明收敛时 $0 < |x_n-A| < \dots < a^{n-1}|x_1-A|$, 而 $\lim_{n \rightarrow \infty} a^{n-1} = 0$. 根据夹逼定理可知... 出收敛时形式时基本公式!

4. 求主部 $k \cdot x^k$ 形式.

① 泰勒公式猜.

② 除以 x^k 洛必达 (eg. 反三角) 直到分母常数

5. 函数的连续性.

① 间断点

$f(x_0), f(x_0)$ 至少一个不存在: 第二类

可去间断点, 不可去间断点, 跳跃间断点

② 闭区间上连续函数的性质.

零点定理

介值定理

有界性定理

最值定理

③ 证明连续: 断点 $f(x_0) = f(x_0)$

证明可导: $\lim_{x \rightarrow x_0} \frac{f(x)-f(x_0)}{x-x_0} = \lim_{x \rightarrow x_0} \frac{f(x)-f(x_0)}{x-x_0}$

连续可导: 证明 $f'(x)$ 连续.

证明无极限: 取固定点 x_0 , 找两个数列

$\{x_n\}, x_n \rightarrow x, \lim_{n \rightarrow \infty} x_n = A$
 $\{y_n\}, y_n \rightarrow x, \lim_{n \rightarrow \infty} y_n = B$
 $\rightarrow A \neq B$. 无极限.

A 级大放缩, 放成 $\frac{1}{n^2}$ 或 $\frac{1}{3^n}$

极限的性质.

1. $\lim_{x \rightarrow x_0} f(x) = A, \lim_{x \rightarrow x_0} g(x) = B$, 则 $A=B$.

如奇数项趋于 A , 偶数项趋于 B , 取 $N = \max\{k_1, k_2\}$ 时 (是 q_0).

2. 有界: 邻域有界, $\lim_{x \rightarrow x_0} f(x) = A \neq \infty$, 则保号性 (邻域)

3. $\lim_{x \rightarrow x_0} f(x) = A \Leftrightarrow \lim_{n \rightarrow \infty} f(x_n) = A$.

4. $\lim x = A \Leftrightarrow x = A + \alpha, \lim \alpha = 0$.

5. 常见的极限: $\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1, \lim_{n \rightarrow \infty} \sqrt[n]{a} = 1 (a > 0)$.

$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \lim_{x \rightarrow \infty} (1 + \frac{1}{x})^x = e$

导数的计算

1. 公式

$(\tan x)' = \frac{1}{\cos^2 x}, (\cot x)' = -\frac{1}{\sin^2 x}$

$(\sec x)' = \frac{\tan x}{\cos^2 x} = \sec x \tan x, (\csc x)' = -\csc x \cot x = -\frac{1}{\sin x} \cdot \frac{1}{\tan x}$

$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}, (\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$

$(\arctan x)' = \frac{1}{x^2+1}, (\text{arccot } x)' = -\frac{1}{x^2+1}$

2. 四则运算

3. 链式求导

4. 幂指函数: 有无穷多平方: 取对数, 拆出根式.

5. 高阶函数: $(\sin x)^{(n)} = \sin(x + \frac{n}{2}\pi)$ 找规律.

$(\cos x)^{(n)} = \cos(x + \frac{n}{2}\pi)$

$(x^\alpha)^{(n)} = \alpha(\alpha-1)\dots(\alpha-n+1)x^{\alpha-n}, (\frac{1}{x})^{(n)} = (-1)^n \frac{n!}{x^{n+1}}$

莱布尼茨公式 $(uv)^{(n)} = u v^{(n)} + C_n^1 u^{(1)} v^{(n-1)} + \dots$

6. 隐函数与参数式函数式.

微分中值定理: 闭区间连续, 开区间可导

$f'(\xi) = \frac{f(b)-f(a)}{b-a}, \frac{f(b)-f(a)}{g(b)-g(a)} = \frac{f'(\xi)}{g'(\xi)}$

$f(x)$ 有三个根 $\Rightarrow f'(x)$ 有两个根.

单箭头

证明不可导:

$\lim_{x_n \rightarrow x_0} \frac{f(x_n)-f(x_0)}{x_n-x_0} = \infty$



期末复习-导数的应用、一元函数微分学.

一、定义域→一阶导→二阶导→单调、凹凸→拐点极值点→渐近线。(作图).

斜渐近线 ≤ 2 , 水平 ≤ 2 , 斜+水平 ≤ 2 .

→eq1. 证明不等式: $(1+a)\ln(1+a) + (1+b)\ln(1+b) < (1+a+b)\ln(1+a+b)$. ($0 < a < b$).

证: 不妨设 $a < b$. 令 $f(x) = (1+x)\ln(1+x)$, $x \in [b, a+b]$. $f'(x) = \ln(1+x) + 1$, $f'(x) > 0$ 在 $(b, a+b)$ 上成立.

由拉格朗日中值定理可知: $f(a+b) - f(b) = a f'(\xi)$, $\xi \in (b, a+b)$.

$f'(x) = \ln(1+x) + 1$, $a f'(\xi) = a \ln(1+\xi) + a$.

只要证 $(1+a)\ln(1+a) < a \ln(1+\xi) + a$.

只要证 $\ln(1+a) < \frac{a}{1+a} \ln(1+\xi) + \frac{a}{1+a}$.

只要证 $(1+a)\ln(1+a) < a \ln(1+\xi) + a$.

只要证 $\ln a < a$, 令 $g(x) = \ln x - x$, $g'(x) = \frac{1}{x} - 1$ ($x > 0$).

$g'(x) = 0$ 时, $x = 1$, $g'(x)$ 在 $(0, 1)$ 上 > 0 , 在 $(1, +\infty)$ 上 < 0 .

$g(x)$ 在 $(0, 1)$ 上 $\uparrow$, 在 $(1, +\infty)$ 上 $\downarrow$, $g(1) = 0$, $g(x) < 0$.

以上各式成立, 证毕.

→eq2. 讨论方程 $ax^3 - 3x^2 + 4 = 0$ ($a > 0$)

按照作图的多项式讨论.

二、不定积分

1. 原函数+常数为不定积分, 是一个集合. (定义)

2. 性质: ① 求导与求积分是互逆运算 (差常数)

$$\textcircled{2} \int (\alpha f(x) + \beta g(x)) dx = \alpha \int f(x) dx + \beta \int g(x) dx.$$

$$\textcircled{3} \int f(u) du = g(v) dv, \text{ 若 } f(u) du = g(v) dv$$

3. 幂指、六个三角, 加:

$$\int \sec^2 x dx, \int \csc^2 x dx, \int \sec x \tan x dx, \int \csc x \cot x dx$$

$$\int \frac{dx}{a^2 - x^2}, \int \frac{dx}{a^2 + x^2}, \int \frac{dx}{x^2 - a^2} \quad (a > 0), \int \frac{dx}{a^2 - x^2}, \int \frac{dx}{x^2 + a^2}$$

$$\int \sqrt{x^2 - a^2} dx, \int \frac{dx}{a^2 + x^2}, \int \frac{dx}{a^2 - x^2}, \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2}$$

$$(a \cos bx + b \sin bx) + C, \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2}$$

$$(a \sin bx - b \cos bx) + C.$$

构成重要不定积分公式.

eg. 求 $\int \frac{x^2 dx}{(x \sin x + \cos x)^2}$

$$\begin{aligned} \text{原式} &= \int \frac{x}{\cos x} \cdot \frac{x \cos x dx}{(x \sin x + \cos x)^2} = \int \frac{x}{\cos x} d \frac{1}{(x \sin x + \cos x)} \\ &= -\frac{x}{\cos x} \cdot \frac{1}{x \sin x + \cos x} - \int -\frac{1}{x \sin x + \cos x} d \left(\frac{x}{\cos x} \right) \\ &= -\frac{x}{x \sin x \cos x + \cos^2 x} + \int \frac{1}{x \sin x + \cos x} \cdot \frac{\cos x + x \sin x}{\cos^2 x} dx \\ &= -\frac{x}{x \sin x \cos x + \cos^2 x} + \tan x + C. \end{aligned}$$

eg. 求 $\int x \arctan^2 x dx$

$$\begin{aligned} \text{原式} &= \arctan^2 x \cdot \frac{x^2+1}{2} - \frac{1}{2} \int (x^2+1) \cdot \frac{2x \arctan x}{x^2+1} dx \\ &= \arctan^2 x \cdot \frac{x^2+1}{2} - \int x \arctan x dx \\ &= \arctan^2 x \cdot \frac{x^2+1}{2} - x \arctan x + \int x \cdot \frac{1}{x^2+1} dx = \arctan^2 x \cdot \frac{x^2+1}{2} - x \arctan x + \frac{1}{2} \ln|x^2+1| + C \end{aligned}$$

eg. $\int \frac{\tan x dx}{1 + \tan x}$

$$\begin{aligned} \text{原式} &= \int \frac{\sin x dx}{\sin x + \cos x} = \frac{1}{2} \int \frac{\sin x + \cos x + \sin x - \cos x}{\sin x + \cos x} dx \\ &= \frac{1}{2} \left(x - \int \frac{d(\sin x + \cos x)}{\sin x + \cos x} \right) = \frac{1}{2} \left(x - \ln|\sin x + \cos x| \right) \\ &= \frac{1}{2} x - \frac{1}{2} \ln|\sin x + \cos x| + C \end{aligned}$$

eg. $\int \frac{1+x^4}{1+x^6} dx$ vs $\int \frac{dx}{x^4+x^6}$

$$\begin{aligned} \int \frac{1+x^4}{1+x^6} dx &= \int \frac{1+x^4-x^4+x^4}{(1+x^2)(1-x^2+x^4)} dx \\ &= \int \frac{dx}{1+x^2} + \int \frac{x^4 dx}{1+x^6} = \arctan x + \frac{1}{3} \arctan(x^3) + C \end{aligned}$$

$$\int \frac{dx}{x^4+x^6} = \int \frac{dx}{x^4(1+x^2)} = \int \frac{(1+x^2)-x^2}{x^4(1+x^2)} dx = \int \frac{dx}{x^4(1+x^2)}$$

$$+ \int \frac{dx}{x^4} = -\frac{1}{3} x^{-3} - \int \frac{(1+x^2)-x^2}{x^4(1+x^2)} dx = -\frac{1}{3x^3} + \frac{1}{x} +$$

$$\arctan x + C$$



积分的定义

eg1. 计算 $I = \lim_{n \rightarrow \infty} \left(\frac{\sin \frac{\pi}{n+1}}{n+1} + \frac{\sin \frac{\pi}{n+2}}{n+2} + \dots + \frac{\sin \frac{\pi}{n}}{n} \right)$

$$I = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{\sin \frac{\pi}{n+k}}{n+k} = \lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{1}{n} \cdot \sin \left(\frac{\pi}{n} \right) \right]$$

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left[\frac{1}{n} \cdot \sin \left(\frac{\pi}{n} \right) \right] = \int_0^1 \sin x dx = -\cos x \Big|_0^1 = -\cos 1 + \cos 0 = 1 - \cos 1$$

由夹逼定理可知, $I = \frac{2}{\pi}$.

识别标志: $\sum_{k=1}^n \frac{1}{n} \rightarrow \int_0^1$ $\lim_{n \rightarrow \infty} \frac{1}{n} \rightarrow 1$

$$f\left(\frac{k}{n}\right) \rightarrow f(x) dx.$$

eg2. 计算 $\lim_{n \rightarrow \infty} \int_0^1 x^n e^x \sin(nx) dx$

法1: $\sin nx$ 为有界量, e^x 为有界量

$\int_0^1 x^n e^x \sin(nx) dx$ 表示 $f(x) = x^n e^x \sin(nx)$ 在 $[0,1]$ 上围成的面积, $0 < \int_0^1 x^n e^x \sin(nx) dx < \int_0^1 x^n e^x dx$

$$\lim_{n \rightarrow \infty} \int_0^1 x^n e^x dx = \lim_{n \rightarrow \infty} \frac{x^{n+1} e^x}{n+1} \Big|_0^1 = \lim_{n \rightarrow \infty} \frac{e}{n+1} = 0.$$

根据夹逼定理, $\lim_{n \rightarrow \infty}$ 原式 $= 0$.

法2: 微分中值定理, 将有界量提出, 原式 $= \lim_{n \rightarrow \infty} e^{\xi} \sin(n\xi) \int_0^1 x^n dx = \frac{e^{\xi} \sin(n\xi)}{n+1} = 0$ ($\xi \in (0,1)$)

若在积分内留其他, 无法“无别量”有界量.

eg3. 设 $f(x) = \int_1^x e^{-t} dt$, 求 $\int_0^1 \frac{f(x)}{x} dx$.

$$\int_0^1 \frac{f(x)}{x} dx = \int_0^1 f(x) d(2\sqrt{x})$$

$$= 2\sqrt{x} f(x) \Big|_0^1 - \int_0^1 2\sqrt{x} d(f(x)) \quad \text{当 } x=0 \text{ 时 } \sqrt{x}=0$$

$$= -2 \int_0^1 \sqrt{x} f'(x) dx \quad \text{当 } x=1 \text{ 时, } f(x)=0.$$

$$= -2 \int_0^1 \sqrt{x} \cdot e^{-x} \cdot \frac{dx}{2\sqrt{x}} = -\int_0^1 e^{-x} dx = e^{-x} \Big|_0^1 = \frac{1}{e} - 1$$

eg4. 设 $f(x) = \int_0^x \frac{\sin t}{\pi-t} dt$, 求 $\int_0^\pi f(x) dx$

$$\int_0^\pi f(x) dx = x f(x) \Big|_0^\pi - \int_0^\pi \frac{x \sin x}{\pi-x} dx$$

$$= \pi f(\pi) - \int_0^\pi \frac{x \sin x}{\pi-x} dx$$

$$= \int_0^\pi \frac{x \sin x}{\pi-x} dx - \int_0^\pi \frac{x \sin x}{\pi-x} dx$$

$$= \int_0^\pi \frac{(x-x) \sin x}{\pi-x} dx = -\cos x \Big|_0^\pi = 2.$$

eg5. 设 $f(x)$ 在 $[1,1]$ 上有阶连续导数, 证明 $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{k^2} f\left(\frac{k}{n}\right) - \int_0^1 f(x) dx$ 存在 (数列极限)

$|f\left(\frac{1}{n}\right) - f(0)| = 2f'(0)$ 存在 (数列极限)

$$\sum_{k=1}^n \frac{1}{k^2} [f\left(\frac{k}{n}\right) - f(0)] = \sum_{k=1}^n \frac{1}{k^2} [f(0) + f'(0)(\frac{k}{n} - 0) + \frac{f''(\xi_k)}{2!} (\frac{k}{n} - 0)^2]$$

$$= f(0) \sum_{k=1}^n \frac{1}{k^2} + \frac{f'(0)}{n} \sum_{k=1}^n \frac{1}{k} + \frac{f''(\xi_k)}{2n^2} \sum_{k=1}^n \frac{1}{k^2}$$

$$= \frac{2}{3} f(0) + \frac{f'(0)}{6n} + \frac{f''(\xi_k)}{6n^2} [2f'(0) + \frac{1}{6n^2} [2f''(\eta)] - 2f'(0)]]$$

$$= \frac{2}{3} f(0) + \frac{f'(0)}{6n} + \frac{f''(\xi_k)}{6n^2} \cdot \frac{M}{3} \cdot \frac{1}{n^2} \leq \frac{M}{3}$$

故 $\sum_{k=1}^n \frac{1}{k^2} f\left(\frac{k}{n}\right) - \int_0^1 f(x) dx$ 存在.

eg6. 设 $f(x)$ 在 $[0,1]$ 连续, 在 $(0,1)$ 可导, $\exists M > 0$, s.t. $|f'(x)| \leq M$, 设 n 是正整数, 证明 $\left| \sum_{k=1}^n \frac{f\left(\frac{k}{n}\right)}{n} - \int_0^1 f(x) dx \right| \leq \frac{M}{2n}$.

$$\left| \sum_{k=1}^n \frac{f\left(\frac{k}{n}\right)}{n} - \sum_{k=1}^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} f(x) dx \right| = \left| \sum_{k=1}^n \left[\frac{f\left(\frac{k}{n}\right)}{n} - \int_{\frac{k-1}{n}}^{\frac{k}{n}} f(x) dx \right] \right| \leq \sum_{k=1}^n \left| \int_{\frac{k-1}{n}}^{\frac{k}{n}} \left| f\left(\frac{k}{n}\right) - f(x) \right| dx \right|$$

$$= \sum_{k=1}^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} |f\left(\frac{k}{n}\right) - f(x)| dx = \sum_{k=1}^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} |f'(\xi)| \left(x - \frac{k-1}{n} \right) dx \leq M \sum_{k=1}^n \int_{\frac{k-1}{n}}^{\frac{k}{n}} \left(x - \frac{k-1}{n} \right) dx = M \sum_{k=1}^n \frac{1}{2} \left(x - \frac{k-1}{n} \right) \Big|_{\frac{k-1}{n}}^{\frac{k}{n}} = \frac{M}{2n}$$

综上所述, 原式 $\leq \frac{M}{2n}$.



eg7. 设 $f(x) \in [a, +\infty)$, 连续且单调减少, $0 < a < b$

证明: $a \int_0^b f(x) dx \leq b \int_0^a f(x) dx$.

设 $g(x) = a \int_0^x f(t) dt - x \int_0^a f(t) dt, x \in (a, b)$.

$$g'(x) = af(x) - \int_0^a f(t) dt, g(a) = 0,$$

$$g(b) = a \int_0^b f(t) dt - b \int_0^a f(t) dt,$$

$$g'(x) = af(x) - af(\xi), \xi \in (a, x).$$

$$= a[f(x) - f(\xi)] \leq 0, g(x) \downarrow, g(b) \leq g(a)$$

即 $g(b) \leq 0$, 代入待证原不等式成立.



定积分在几何上的应用.

微元法: 平面图形面积的计算公式.
已知横截面面积的空间图形体积.
曲线的弧长.

eg1. 设 $f(x)$ 定义域与值域均为 $[0,1]$, $f(x)$ 是一阶连续函数, $f'(x)$ 是单调减函数, $f(0)=f(1)=0$, 且 $y=f(x)$, $x \in [0,1]$, f 所围成的曲线弧长不超过 3, $S = \int_0^1 \sqrt{1+f'(x)^2} dx$.

$$S = \int_0^1 \sqrt{1+f'(x)^2} dx, \text{ 而 } f(0)=f(1)=0, \text{ 故 } \exists \xi \in (0,1) \text{ 使 } f'(\xi)=0, \text{ 在 } (0,\xi) \text{ 上 } f'(x) > 0, \text{ 在 } (\xi,1) \text{ 上 } f'(x) < 0, \text{ 故 } S = \int_0^\xi \sqrt{1+f'(x)^2} dx + \int_\xi^1 \sqrt{1+f'(x)^2} dx = \int_0^\xi (f'(x)+1) dx + \int_\xi^1 (-f'(x)+1) dx = f(\xi) + \xi + 1 - f(\xi) = 1 + \xi + f(\xi).$$

$$S(\xi,1) \leq \int_\xi^1 \sqrt{1+f'(x)^2} dx \leq \int_\xi^1 (1+f'(x)) dx = 1 - f(\xi) = 1 - f(1) = 1. \text{ 故 } S \leq 3.$$

$$S = S(0,\xi) + S(\xi,1) = 1 + 2f(\xi), f(\xi) \in [0,1], \text{ 故 } S \leq 3.$$

广义积分.

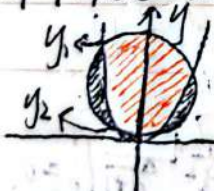
无穷分为奇点, 注意断点.

eg2. $\int_0^3 \frac{x}{\sqrt{(x-1)(3-x)}} dx$.

$$\int_0^3 \frac{x}{\sqrt{(x-1)(3-x)}} dx = \int_{-1}^1 \frac{t+1}{\sqrt{1-t^2}} dt = 2 \int_{-1}^1 \frac{t}{\sqrt{1-t^2}} dt + \int_{-1}^1 \frac{1}{\sqrt{1-t^2}} dt = 2 \int_{-1}^1 \frac{t}{\sqrt{1-t^2}} dt = 2 \arcsin t \Big|_{-1}^1 = 2(\frac{\pi}{2} - \frac{\pi}{2}) = 2\pi. \text{ 注意: 分母不可为0, 故去掉奇点.}$$

空间解析几何

eg1. 设位于圆内部 $x^2 + (y-a)^2 \leq c^2$ ($a > c > 0$) 内部而在抛物线 $x^2 = 2by$ ($a > b > 0, c^2 > 2ab - b^2$) 外部的部分区域为 D , 求 D 绕 y 轴旋转一周所得旋转体的体积.



$$V = \text{球} - \text{橙色} = \pi \int_{y_2}^{y_1} [c^2 - (y-a)^2 - 2by] dy = \pi \int_{y_2}^{y_1} (-y^2 + 2ay + c^2 - a^2) dy = \pi \left(-\frac{1}{3}y^3 + (a+\frac{c^2-a^2}{2})y^2 \right) \Big|_{y_2}^{y_1} = \frac{\pi}{3} (y_1 - y_2) [(y_1 + y_2)^2 - y_1 y_2] + \pi(a-b)(y_1 - y_2)(y_1 + y_2) + (c^2 - a^2)(y_1 - y_2).$$

$$c^2 - (y-a)^2 = 2by.$$

$$y^2 - 2(a+b)y + c^2 = a^2.$$

$$(y-a)^2 + 2by = c^2.$$

$$y^2 - 2(a+b)y - (c^2 - a^2) = 0.$$

$$y^2 - 2ay + a^2 + 2by = c^2.$$

$$y_1 + y_2 = 2(a+b), y_1 y_2 = a^2 - c^2.$$

$$y_1 - y_2 = \sqrt{(y_1 + y_2)^2 - 4y_1 y_2} = \sqrt{4a^2 + 4b^2 - 8ab - 4a^2 + 4c^2} = 2\sqrt{b^2 + c^2 - 2ab}.$$

$$\text{求 } = 2\sqrt{b^2 + c^2 - 2ab}.$$

eg2. 由曲线 $y = \ln x$ 在 $(e,1)$ 处的切线 $y = \frac{1}{e}x$ 以及 x 轴所围成的平面图形 D 的面积 S .

D 分别绕 x 轴, y 轴旋转一周所得旋转体体积为 V_x, V_y .

$$y = \frac{1}{e}, \text{ 切线 } l: y = \frac{1}{e}(x-e) + 1 = \frac{1}{e}x. \quad S = \int_0^e \frac{1}{e}x dx + \int_e^e (\frac{1}{e}x - \ln x) dx = \frac{1}{2e}x^2 \Big|_0^e + \frac{1}{e}x^2 \Big|_e^e - \int_e^e \ln x dx = \frac{1}{2e}e^2 - \int_e^e \ln x dx = \frac{e}{2} - (x \ln x - x) \Big|_e^e = \frac{e}{2} - (e - e) + (e - 1) = \frac{e}{2} - 1.$$

$$\int \ln x dx = x \ln x - \int 1 dx = x \ln x - x.$$

$$V_x = \pi \int_0^e (\frac{1}{e}x)^2 dx + \pi \int_e^e (\frac{1}{e}x - \ln x)^2 dx = \pi \int_0^e (\frac{1}{e}x)^2 dx = \frac{\pi}{3e^3}x^3 \Big|_0^e = \frac{\pi e^3}{3}.$$

$$V_y = 2\pi \int_0^e x(\frac{1}{e}x) dx + 2\pi \int_e^e x(\frac{1}{e}x - \ln x) dx = \frac{\pi e^3}{2} + \frac{\pi e^3}{2} = \pi e^3.$$



$$\text{eq1. } \int \frac{dx}{(x^2-x-2)(x-1)^2}$$

$$f(x) = \int \frac{dx}{(x-2)(x+1)(x-1)^2}$$

$$\hat{=} g(x) = \frac{dx}{(x-2)(x+1)(x-1)^2}$$

$$= \frac{A}{x-2} + \frac{B}{x+1} + \frac{C}{x-1} + \frac{D}{(x-1)^2}$$

$$\star g(x)(x-2)|_{x=2} = A = \frac{1}{(x+1)(x-1)^2} = \frac{1}{3}$$

$$g(x)(x+1)|_{x=-1} = B = \frac{1}{(x-2)(x-1)^2} = -\frac{1}{12}$$

$$g(x)(x-1)^2|_{x=1} = D = \frac{1}{(x+1)(x-2)} = -\frac{1}{2}$$

$$[g(x)(x-1)^2]'|_{x=1} = E = \frac{-2(x-1)}{(x+1)(x-2)^2} = -\frac{1}{4}$$

$$f(x) = \frac{1}{3} \int \frac{dx}{x-2} - \frac{1}{12} \int \frac{dx}{x+1} - \frac{1}{4} \int \frac{dx}{x-1} - \frac{1}{2} \int \frac{dx}{(x-1)^2}$$

$$= \frac{1}{3} \ln|x-2| - \frac{1}{12} \ln|x+1| - \frac{1}{4} \ln|x-1| + \frac{1}{2(x-1)} + C$$

$$\text{eq2. } \int \frac{dx}{x^2+1}$$

$$\text{原式} = \frac{1}{2} \int \frac{(x^2+1) - (x^2-1)}{x^2+1} dx$$

$$= \frac{1}{2} \int \frac{(1+\frac{1}{x^2})}{x^2+1} dx - \frac{1}{2} \int \frac{(1-\frac{1}{x^2})}{x^2+1} dx$$

$$= \frac{1}{2} \int \frac{d(x+\frac{1}{x})}{(x+\frac{1}{x})^2+2} - \frac{1}{2} \int \frac{d(x+\frac{1}{x})}{(x+\frac{1}{x})^2-2}$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \arctan\left(\frac{x+\frac{1}{x}}{\sqrt{2}}\right) - \frac{1}{2} \int \frac{d(x+\frac{1}{x})}{(x+\frac{1}{x})^2-2} + C_1$$

$$\hat{=} u = x + \frac{1}{x}, \int \frac{du}{u^2-2} = \frac{1}{2\sqrt{2}} \left(\int \frac{d(u+\sqrt{2})}{u+\sqrt{2}} - \int \frac{d(u-\sqrt{2})}{u-\sqrt{2}} \right)$$

$$= \frac{1}{2\sqrt{2}} \ln \left| \frac{u+\sqrt{2}}{u-\sqrt{2}} \right| + C_2$$

$$\text{原式} = \frac{\sqrt{2}}{4} \arctan\left(\frac{x+\frac{1}{x}}{\sqrt{2}}\right) + \frac{\sqrt{2}}{4} \ln \left| \frac{x^2\sqrt{2}+1}{x^2\sqrt{2}-1} \right| + C$$

$$\text{eq3. } \int \frac{dx}{x\sqrt{x^2+5x+1}}$$

$$\text{设 } t = \frac{1}{x}, \text{原式} = \int \frac{d(\frac{1}{t})}{\frac{1}{t}(\frac{1}{t^2}+5\frac{1}{t}+1)} = \int \frac{-\frac{1}{t^2} dt}{\frac{1}{t^3} + 5\frac{1}{t^2} + 1} dt$$

$$= \int \frac{dt}{\frac{1}{t^3} + 5\frac{1}{t^2} + 1} = \int \frac{dt}{\frac{1+t^2+5t}{t^3}} = \int \frac{t^3 dt}{1+t^2+5t}, \hat{=} t + \frac{5}{2} = u$$

$$= \int \frac{\frac{2}{3} du}{\frac{1}{4} u^2 - \frac{25}{4}} = \ln \left| \frac{u + \frac{5}{2}}{u - \frac{5}{2}} \right| = \ln \left| \frac{t + \frac{5}{2}}{t - \frac{5}{2}} \right| = \ln \left| \frac{1 + \frac{5}{2}t}{1 - \frac{5}{2}t} \right|$$

$$= \ln \left| \frac{1 + \frac{5}{2}t}{1 - \frac{5}{2}t} \right| = \ln \left| \frac{1 + \frac{5}{2} \cdot \frac{1}{x}}{1 - \frac{5}{2} \cdot \frac{1}{x}} \right| = \ln \left| \frac{x + \frac{5}{2}}{x - \frac{5}{2}} \right|$$

根号在底下的，且5-2次方被约掉，所以列代换降次

$$\text{eq3. } \int \frac{\tan x dx}{a^2 \sin^2 x + b^2 \cos^2 x}$$

$$\text{原式} = \int \frac{\tan x d(\tan x)}{a^2 \tan^2 x + b^2}, \hat{=} u = \tan x$$

$$= \int \frac{u du}{a^2 u^2 + b^2} = \frac{1}{2} \int \frac{du}{a^2 u^2 + b^2} = \frac{1}{2a^2} \int \frac{d(a^2 u^2 + b^2)}{a^2 u^2 + b^2}$$

$$= \frac{1}{2a^2} \ln|a^2 u^2 + b^2| + C_1$$

$$= \frac{1}{2a^2} \ln|a^2 \tan^2 x + b^2| + C$$

$$\text{eq4. } \int \frac{\sin x}{1+\sin x} dx$$

$$\text{法一: } I = \int \frac{\sin x}{1+\sin x} dx = \int \frac{dx}{1+\sin x} = \int \frac{1-\sin x}{\cos^2 x} dx$$

$$= \int \frac{dx}{\cos^2 x} - \int \frac{\sin x}{\cos^2 x} dx = \int \sec^2 x dx - \int \tan x \sec x dx$$

$$\text{法二: } I = \int \frac{\sin(x+\frac{\pi}{2})}{1+\sin(x+\frac{\pi}{2})} d(x+\frac{\pi}{2}) = \int \frac{1}{1+\sin(x+\frac{\pi}{2})} d(x+\frac{\pi}{2})$$

$$= \int \frac{1}{1+\cos x} dx = \int \frac{dx}{2\cos^2(\frac{x}{2})} = \int \frac{d(\frac{x}{2})}{\cos^2(\frac{x}{2})}$$

$$= \int \sec^2(\frac{x}{2}) d(\frac{x}{2}) = \tan(\frac{x}{2}) + C$$

$$\text{eq5. } \int \frac{dx}{\sin 2x + 2\sin x}$$

$$I = \int \frac{dx}{2\sin x \cos x + 2\sin x} = \int \frac{dx}{2\sin x (\cos x + 1)}$$

$$= \frac{1}{4} \int \frac{d(\frac{x}{2})}{\cos(\frac{x}{2}) \sin(\frac{x}{2})} = \frac{1}{4} \int \frac{d(\tan(\frac{x}{2}))}{\tan(\frac{x}{2}) \cos^2(\frac{x}{2})}, \frac{x}{2} = u$$

$$= \frac{1}{4} \int \frac{d(\tan u)}{\tan u \cdot \tan^2 u + 1} = \frac{1}{4} \int \frac{d(\tan u)}{\tan^3 u + \tan u} = \frac{1}{4} \left(\frac{1}{2} \tan^2(\frac{x}{2}) + \ln|\tan(\frac{x}{2})| \right) + C$$

法二: 万能代换

$$\text{eq6. } \lim_{n \rightarrow \infty} \int_0^1 \frac{x^n}{1+x^n} dx$$

$$I = \lim_{n \rightarrow \infty} \frac{1}{1+\frac{1}{n^2}} \int_0^1 x^n dx = \lim_{n \rightarrow \infty} \frac{1}{1+\frac{1}{n^2}} \cdot \frac{x^{n+1}}{n+1} \Big|_0^1 = 0$$

$$\text{eq7. } \int \frac{1+x}{x(1-xe^x)} dx$$

$$I = \int \frac{(1+x)e^x}{x e^x (1-xe^x)} dx = \int \frac{d(xe^x)}{xe^x (1-xe^x)}, \hat{=} t = xe^x$$

$$I = \int \frac{dt}{t(1-t)} = \int \frac{dt}{t} + \int \frac{dt}{1-t} = \ln|t| - \ln|1-t| + C_0$$

$$= \ln \left| \frac{xe^x}{1-xe^x} \right| + C$$

$$\text{eq8. } \int \frac{x^2}{1+x^2} dx$$

$$I = \int \frac{x^2}{1+x^2} d(x^2+1) = \int \frac{x^2+1-1}{1+x^2} d(x^2+1)$$

$$\hat{=} t = x^2+1, I = \int \frac{t-1}{t} dt = \int \left(1 - \frac{1}{t} \right) dt$$

$$= \frac{1}{3} t^{\frac{3}{2}} - t + C_0 = \frac{1}{3} (x^2+1)^{\frac{3}{2}} - (x^2+1) + C$$



$$\begin{aligned} \text{eg 9. } \lim_{x \rightarrow 0} \frac{1}{\sin(x)} \int_0^x \frac{e^t - 1}{t} dt &= \lim_{x \rightarrow 0} \frac{\int_0^x \frac{e^t - 1}{t} dt}{\sin(x)} \\ I &= \lim_{x \rightarrow 0} \frac{\int_0^x \frac{e^t - 1}{t} dt}{\sin(x)} = \lim_{x \rightarrow 0} \frac{\int_0^x \frac{e^t - 1}{t} dt}{x^2} \\ \int_0^x \frac{e^t - 1}{t} dt &= \int_0^x \frac{e^u - 1}{u} du \\ I &= \lim_{x \rightarrow 0} \frac{e^x - 1}{2x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x^2} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{eg 9. } \int_0^{\frac{\pi}{2}} \ln(1 + \tan x) dx &= \int_0^{\frac{\pi}{2}} \ln\left(\frac{\sin x + \cos x}{\cos x}\right) dx \\ &= \int_0^{\frac{\pi}{2}} \ln(\sin x + \cos x) dx - \int_0^{\frac{\pi}{2}} \ln(\cos x) dx \\ &= \left[\frac{\pi}{4} \ln(2) + \int_0^{\frac{\pi}{2}} \ln(\cos(\frac{\pi}{2} - x)) dx \right] - \int_0^{\frac{\pi}{2}} \ln(\cos x) dx \\ &= \int_0^{\frac{\pi}{2}} \ln(\cos x) dx = \frac{\pi}{8} \ln 2 + \int_0^{\frac{\pi}{2}} \ln(\cos u) du \\ &+ \int_0^{\frac{\pi}{2}} \ln(\cos x) dx = \frac{\pi}{8} \ln 2. \end{aligned}$$

$$\begin{aligned} \text{eg 10. } \int_0^1 \frac{\ln(x+1)}{x^2+1} dx &= \int_0^1 \ln(x+1) d(\arctan x) \\ &= \frac{\pi}{4} \ln 2 - \int_0^1 \frac{1}{x^2+1} \arctan x d(\ln(x+1)) \\ &= \frac{\pi}{4} \ln 2 - \int_0^1 \frac{1}{x^2+1} \arctan x dx, \text{ let } t = \arctan x \\ &= \frac{\pi}{4} \ln 2 - \int_0^{\frac{\pi}{4}} \frac{1}{\tan^2 t + 1} t d(\tan t) \\ &= \frac{\pi}{4} \ln 2 - \int_0^{\frac{\pi}{4}} \frac{t dt}{\cos^2 t (\sin t + \cos t)}, \text{ let } u = \frac{\pi}{4} - t \\ &= \frac{\pi}{4} \ln 2 - \int_0^{\frac{\pi}{4}} \frac{(\frac{\pi}{4} - u) du}{\cos u (\sin u + \cos u)} \\ &= \frac{\pi}{4} \ln 2 - \frac{\pi}{8} \int_0^{\frac{\pi}{4}} \frac{d(\tan u)}{1 + \tan u} + \int_0^{\frac{\pi}{4}} \frac{u du}{\cos u (\sin u + \cos u)} \\ &= \frac{\pi}{8} \ln 2 - \frac{\pi}{8} \int_0^{\frac{\pi}{4}} \frac{d(\tan u)}{1 + \tan u}, \int_0^{\frac{\pi}{4}} \frac{d(\tan u)}{1 + \tan u} \\ &= \frac{\pi}{8} \ln 2 - \frac{\pi}{8} \ln 2 = \frac{\pi}{8} \ln 2. \end{aligned}$$

$$\begin{aligned} \text{eg 11. } \int_0^1 \frac{x^9 dx}{(x^{10} + 2x^5 + 2)^2} &= \int_0^1 \frac{1}{10} \frac{d(x^{10})}{(x^{10} + 2x^5 + 2)^2}, \text{ let } x^5 = t, \frac{1}{10} \frac{d(t^2)}{(t^2 + 2t + 2)^2} \\ &= \frac{1}{5} \int_0^1 \frac{t dt}{(t^2 + 2t + 2)^2}, \text{ let } t+1 = \tan t, t = \tan t - 1 \\ &= \frac{1}{5} \int_0^1 \frac{(\tan t - 1) \sec^2 t dt}{\sec^4 t} = \frac{1}{5} \int_0^1 \left(\frac{\tan t}{\sec^2 t} - \frac{1}{\sec^2 t} \right) dt \\ &= \frac{1}{5} \int_0^1 \sin t \cos t dt - \frac{1}{5} \int_0^1 \cos^2 t dt \\ &= \frac{1}{20} \int_0^1 \sin 2t d(2t) - \frac{1}{20} \int_0^1 (\cos 2t + 1) d(\cos 2t) \\ &= -\frac{1}{20} \cos 2t - \frac{1}{20} \sin 2t - \frac{1}{10} + C. \text{ 换回原变量} \\ &= -\frac{1}{20} (\cos 2t + \sin 2t) \end{aligned}$$

$$\begin{aligned} \text{eg 11. } \text{原式} &= \frac{1}{5} \int_0^1 \frac{t dt}{(t^2 + 2t + 2)^2} = \frac{1}{10} \int_0^1 \frac{2t + 2 - 2}{(t^2 + 2t + 2)^2} dt \\ &= \frac{1}{10} \int_0^1 \frac{d(t^2 + 2t + 2)}{(t^2 + 2t + 2)^2} - \frac{1}{5} \int_0^1 \frac{d(t+1)}{(t+1)^2} \\ &= \frac{1}{10} \left(-\frac{1}{t^2 + 2t + 2} \right) - \frac{1}{5} \left(-\frac{1}{t+1} \right) + C_0 \\ &= -\frac{1}{10(t^2 + 2t + 2)} + \frac{1}{5(t+1)} + \tan(\arctan(t+1)) \\ &= -\frac{1}{10(x^{10} + 2x^5 + 2)} + \frac{1}{5(x^5 + 1)} + \tan(\arctan(x^5 + 1)) \\ &+ C \end{aligned}$$

eg 12. $y = 3x + 2$ 与 $y = x^3$ 相交构成的部分为 D , 求 $x > 0$ 部分绕 x 轴旋转一周所得旋转体体积 V

即题中所给直线与 x 轴围成的部分.

$$\begin{aligned} y &= 3x + 2, \text{ 相交解得 } x = -1 \text{ 或 } x = 2. \\ y &= x^3 \\ V &= \pi \int_0^2 (3x+2)^2 dx - \pi \int_0^2 (x^3)^2 dx \\ &= \pi \cdot 9 \cdot \frac{1}{3} x^3 \Big|_0^2 + \pi \cdot 12 \cdot \frac{1}{2} x^2 \Big|_0^2 + 4\pi x \Big|_0^2 - \frac{1}{7} \pi x^7 \Big|_0^2 \\ &= 3\pi \cdot 8 + 6\pi \cdot 4 + 8\pi - \frac{128}{7} \pi = \frac{264}{7} \pi. \end{aligned}$$



eg13 设 $f(x)$ 在 $[0,1]$ 连续, 且 $f(x) \geq 0$, 满足 $f(x) \leq 1 + 2 \int_0^x f(t) dt$, $x \in [0,1]$, 求证: $f(x) \leq 1+x$, $x \in [0,1]$.

设 $F(x) = 1 + 2 \int_0^x f(t) dt$, $F(x) \geq 0$

$$F'(x) = 2f(x) \geq 0, F(0) = 1.$$

$$F(x) = 1 + 2 \int_0^x f(t) dt \leq 2\sqrt{F(x)}, f(x) \leq \sqrt{F(x)}$$

$$\sqrt{F(x)} - 1 = \frac{2 \int_0^x f(t) dt}{\sqrt{F(x)}} \quad \text{对 } \sqrt{F(x)} - 1 \text{ 求导:}$$

$$\sqrt{F(x)} - 1 = \int_0^x \frac{F'(t)}{2\sqrt{F(t)}} dt = \int_0^x \frac{2f(t)}{2\sqrt{F(t)}} dt \leq \int_0^x 1 dt = x, \sqrt{F(x)} - 1 \leq x, \sqrt{F(x)} \leq x+1.$$

$$f(x) \leq \sqrt{F(x)} \leq x+1, \text{ 证毕.}$$

本题在可导没有控制导数连续因此不用莱布尼茨, 利用 $(\int_0^x f(t) dt)' = f(x)$

eg14. $f(x)$ 在 $[a,a]$ 上有二阶导数.

$$(1) f''(x) > 0, (x \in [a,a]), \text{ 证: } \int_{-a}^a f(x) dx \geq 2af(0)$$

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2} f''(\xi) \geq f(0) + xf'(0)$$

$$\int_{-a}^a f(x) dx \geq \int_{-a}^a [f(0) + xf'(0)] dx \quad \text{奇函数.}$$

$$\geq 2af(0) + \frac{1}{2} x^2 f''(0) \Big|_{-a}^a = 0$$

$$\geq 2af(0) \quad \text{原不等式得证}$$

(2) 若 $f(0) = 0$, 证: 在 $[a,a]$ 上至少存在一点 ξ ,

$$\text{使 } a^3 f''(\xi) = 3 \int_{-a}^a f(x) dx$$

$$F(x) = \int_0^x f(t) dt, F(x) = F(0) + xF'(0) + \frac{x^2}{2} F''(0) + \frac{x^3}{6} F'''(\xi)$$

$$F(a) = F(0) + a f(0) + \frac{a^2}{2} f'(0) + \frac{a^3}{6} f''(\xi_1), F(-a) = F(0) + (-a) f(0) + \frac{a^2}{2} f'(0) - \frac{a^3}{6} f''(\xi_2)$$

$$\text{① } F(0) = 0, f(0) = 0, \text{ 故 } F(a) = \frac{a^2}{2} f'(0) + \frac{a^3}{6} f''(\xi_1), \text{ ② } F(-a) = \frac{a^2}{2} f'(0) - \frac{a^3}{6} f''(\xi_2)$$

$$\text{③ } F(a) - F(-a) = \frac{a^3}{3} f''(\xi) = \frac{a^3}{3} (f''(\xi_1) + f''(\xi_2)) = \frac{a^3}{3} (f''(\xi_1) + f''(\xi_2))$$

$$\text{④ } F(a) - F(-a) = \frac{a^3}{3} f''(\xi) = \frac{a^3}{3} (f''(\xi_1) + f''(\xi_2)) = \frac{a^3}{3} (f''(\xi_1) + f''(\xi_2))$$

$$\text{⑤ } F(a) - F(-a) = \frac{a^3}{3} f''(\xi) = \frac{a^3}{3} (f''(\xi_1) + f''(\xi_2)) = \frac{a^3}{3} (f''(\xi_1) + f''(\xi_2))$$

$$\text{⑥ } F(a) - F(-a) = \frac{a^3}{3} f''(\xi) = \frac{a^3}{3} (f''(\xi_1) + f''(\xi_2)) = \frac{a^3}{3} (f''(\xi_1) + f''(\xi_2))$$

$$\text{⑦ } F(a) - F(-a) = \frac{a^3}{3} f''(\xi) = \frac{a^3}{3} (f''(\xi_1) + f''(\xi_2)) = \frac{a^3}{3} (f''(\xi_1) + f''(\xi_2))$$

$$= \frac{a^3}{3} f''(\xi), 3 \int_{-a}^a f(x) dx = a^3 f''(\xi).$$



期末复习：证明题

1. 换元法

eg1: 设 $f(x)$ 在 $\mathbb{R}$ 上连续, 且 $F(x) = \int_0^x (x-t)f(t)dt$

证明: (1) 若 $f(x)$ 为偶函数, 则 $F(x)$ 也为偶函数.

证: $F(-x) = \int_0^{-x} (-x-t)f(t)dt$, 令 $u = -t$

$$\begin{aligned} F(-x) &= \int_0^{-x} (-x-t)f(t)dt = -\int_0^x (-x+u)f(-u)du \\ &= \int_0^x (x-u)f(u)du = \int_0^x (x-t)f(t)dt \\ &= F(x), \text{ 定义域为 } \mathbb{R} \text{ (关于原点对称)}. \end{aligned}$$

(2) 若 $f(x)$ 为奇函数, 则 $F(x)$ 为偶函数 (微分中值定理).

证: $F(x) = x \int_0^x f(t)dt - 2 \int_0^x tf(t)dt$

$$F'(x) = \int_0^x f(t)dt + x f(x) - 2x f(x)$$

$$= \int_0^x f(t)dt - x f(x) = x f(\xi) - x f(x)$$

$$= x[f(\xi) - f(x)], \quad x > 0 \text{ 时, } \xi \in (0, x), f(\xi) > f(x)$$

$x < 0$ 时, $\xi \in (x, 0), f(\xi) < f(x), F'(x) > 0, F(x) \uparrow$

eg2: $f(x)$ 为 $[a, a]$ 的偶函数, 连续, 证明:

$$\int_{-a}^a \frac{f(x)}{1+e^x} dx = \int_0^a f(x) dx$$

$$\begin{aligned} \text{证: 设 } & \int_0^a \frac{f(x)}{1+e^x} dx + \int_{-a}^0 \frac{f(x)}{1+e^x} dx = \int_0^a \frac{f(x)}{1+e^x} dx + \int_0^a \frac{f(-x)}{1+e^{-x}} dx \\ &= \int_0^a \frac{f(x)}{1+e^x} dx + \int_0^a \frac{f(x)}{1+e^x} dx = \int_0^a f(x) dx \\ & \left(\frac{1}{1+e^x} + \frac{1}{1+e^{-x}} \right) dx = \int_0^a f(x) dx, \text{ 原式证毕.} \end{aligned}$$

识别标志: 奇偶性.

2. 构造函数法

eg3: $f(x)$ 在 $[a, b]$ 连续, 且 $\int_a^b f(x) dx = \int_a^b x f(x) dx \neq 0$

证明 $f(x)$ 在 (a, b) 内至少有一个零点, (中值定理)

令 $F(x) = \int_a^x f(t)dt, F(a) = 0, F(b) = 0$

$$\begin{aligned} \int_a^b x f(x) dx &= \int_a^b x d(F(x)) = x F(x) \Big|_a^b - \int_a^b F(x) dx \\ &= - \int_a^b F(x) dx = 0, \int_a^b F(x) dx = 0, F(\xi)(b-a) = 0 \end{aligned}$$

$b-a \neq 0, F(\xi) = 0$, 在 $(a, \xi), F(a) - F(\xi) = (a-\xi)f(\eta_1) = 0$,

$F(b) - F(\xi) = (b-\xi)f(\eta_2) = 0$, 故至少有一个 $f(\eta_1) = f(\eta_2) = 0$

识别标志: 函数连续, 说明可能要用到

原函数和不定积分, 如 $f(\xi)(b-a) = F(b) - F(a)$

或 $\int_a^b f(x) dx = f(\xi)(b-a)$

eg4: $f \in C^1([0, 1])$, 求证 $\forall x \in [0, 1]$ 有 $|f(x)| \leq \int_0^1 (|f(t)| + |f'(t)|) dt$

只要证 $|f(x)| \leq \int_0^1 |f(t)| dt + \int_0^1 |f'(t)| dt$

$$f(x) = f(0) + \int_0^x f'(t) dt$$

$f \in C^1([0, 1]) \Rightarrow f(0) = \int_0^1 f(t) dt$, 由牛顿-莱布尼兹 (取 $\int_0^1 f(t) dt = f(\xi) \int_0^1 1 dt = f(\xi)$) 定理可知:

$$\begin{aligned} |f(x) - f(\xi)| &= \left| \int_{\xi}^x f'(t) dt \right|, |f(x)| = |f(\xi) + \int_{\xi}^x f'(t) dt| \\ &\leq |f(\xi)| + \int_{\xi}^x |f'(t)| dt \leq \int_0^1 |f(t)| dt + \int_{\xi}^x |f'(t)| dt \\ &\leq \int_0^1 |f(t)| dt + \int_0^1 |f'(t)| dt = \int_0^1 (|f(t)| + |f'(t)|) dt. \end{aligned}$$

or: $\int_0^1 (\int_0^1 |f'(s)| ds) dt$, 令常数 $M = \int_0^1 |f'(s)| ds$

$$\int_0^1 M dt = M = \int_0^1 |f'(s)| ds = \int_0^1 |f'(t)| dt$$

eg5: $f(x), g(x)$ 在 $[a, b]$ 上连续, 求证: $\exists \xi \in (a, b)$

s.t. $f(\xi) \int_a^b g(x) dx = g(\xi) \int_a^b f(x) dx$ (罗尔定理)

设 $F(x) = \int_a^x f(t) dt, G(x) = \int_a^x g(t) dt, F(a) = F(b) = 0$

$$\therefore F(a) - F(b) = F(\xi)(a-b) = 0, \text{ 且 } F'(\xi) = 0$$

$$F'(x) = f(x), G'(x) = g(x), F(\xi) = 0, G(\xi) = 0$$

$$F(\xi) = -f(\xi) \int_a^{\xi} g(x) dx + g(\xi) \int_a^{\xi} f(x) dx = 0$$

即原式成立, 证毕.



eg. 6. 设 $f(x)$ 在 $[a, b]$ 上单调增加, 连续, 求证:

$$\int_a^b x f(x) dx \geq \frac{a+b}{2} \int_a^b f(x) dx.$$

$$\text{证 } F(x) = \int_a^x x f(x) dx - \frac{a+x}{2} \int_a^x f(x) dx, F(a) = 0.$$

$$F'(x) = x f(x) - \frac{a}{2} f(x) - \frac{1}{2} \int_a^x f(x) dx - \frac{x}{2} f(x)$$

$$= \frac{1}{2}(x-a)f(x) - \frac{1}{2} \int_a^x f(x) dx, F(a) = 0$$

$$F''(x) = \frac{1}{2}f(x) + \frac{1}{2}f'(x)(x-a) - \frac{1}{2}f(x) = \frac{1}{2}f'(x)(x-a)$$

$$x \in [a, b], f'(x) \geq 0, F''(x) \geq 0, F'(x) \downarrow, F'(x) \geq 0,$$

$$F'(x) \uparrow, F(b) \geq F(a) = 0, \text{原式证毕}$$

3. 泰勒公式, 施瓦茨不等式.

$$\text{施瓦茨不等式: } \left(\int_a^b g(x) f(x) dx \right)^2 \leq \left(\int_a^b g^2(x) dx \right) \left(\int_a^b f^2(x) dx \right).$$

eg 7. 设 $f(x)$ 在 $[a, b]$ 上连续可导, $f(a) = 0$, 求:

$$\int_a^b f^2(x) dx \leq \frac{(b-a)^2}{2} \int_a^b f''(x) dx. \text{ (施瓦茨)}$$

$$f(x) = f(x) - f(a) = \int_a^x f'(x) dx, x \in [a, b]$$

$$f^2(x) = \left(\int_a^x f'(x) dx \right)^2 \leq \left(\int_a^x 1^2 dx \right) \left(\int_a^x f'^2(x) dx \right)$$

$$= (x-a) \left(\int_a^x f'^2(x) dx \right) \leq (x-a) \left(\int_a^b f'^2(x) dx \right)$$

$$\int_a^b f^2(x) \leq \int_a^b (x-a) dx \cdot \int_a^b f'^2(x) dx$$

$$= \frac{(b-a)^2}{2} \left(\int_a^b f'^2(x) dx \right)$$

$$\therefore \int_a^b f^2(x) \leq \frac{(b-a)^2}{2} \left(\int_a^b f'^2(x) dx \right)$$

eg 8. $f(x)$ 在 $[a, b]$ 上二阶连续可导, 求证:

$$\int_a^b f(x) dx \leq (b-a) f\left(\frac{a+b}{2}\right) + \frac{1}{24}(b-a)^3 \max_{a \leq x \leq b} f''(x)$$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(\xi)}{2}(x-x_0)^2$$

$$\int_a^b f(x) = \int_a^b f(x_0) dx + f'(x_0) \int_a^b (x-x_0) dx + \frac{f''(\xi)}{2} \int_a^b (x-x_0)^2 dx$$

$$(x-x_0) d(x-x_0) = f\left(\frac{a+b}{2}\right)(b-a) + f'(x_0) \cdot \frac{1}{2}(x-\frac{a+b}{2})^2 \Big|_a^b$$

$$+ \frac{f''(\xi)}{2} \cdot \frac{1}{3} (x-\frac{a+b}{2})^3 \Big|_a^b = f\left(\frac{a+b}{2}\right)(b-a) + \frac{1}{24}(b-a)^3 f''(\xi)$$

$$\leq f\left(\frac{a+b}{2}\right)(b-a) + \frac{1}{24}(b-a)^3 \max_{x \in [a, b]} f''(x). \text{ 证毕}$$

eg 9. $f(x)$ 在 $[0, 2]$ 上具有二阶连续导数:

$$(1) \text{ 如果 } f(1) = 0, \text{ 证明: } \left| \int_0^2 f(x) dx \right| \leq \frac{1}{3} \max_{x \in [0, 2]} |f''(x)|$$

$$f(x) = f(1) + f'(1)(x-1) + \frac{f''(\xi)}{2}(x-1)^2 = f'(1)(x-1) + \frac{f''(\xi)}{2}(x-1)^2$$

$$\int_0^2 f(x) dx = \int_0^2 f'(1)(x-1) dx + \int_0^2 \frac{f''(\xi)}{2}(x-1)^2 dx$$

$$= \frac{1}{2}(x-1)^2 f'(1) \Big|_0^2 + \frac{f''(\xi)}{6}(x-1)^3 \Big|_0^2 = \frac{f''(\xi)}{6}(x-1)^3 \Big|_0^2$$

$$\leq \frac{1}{3} \max_{x \in [0, 2]} |f''(x)|$$

$$\text{绝对值: } \left| \int_0^2 f(x) dx \right| = \left| \int_0^2 \frac{f''(\xi)}{2}(x-1) dx \right| \leq \int_0^2 \left| \frac{f''(\xi)}{2}(x-1) \right| dx$$

$$= \int_0^2 \left| \frac{f''(\xi)}{2} \right| |x-1| dx = \int_0^2 \left| \frac{f''(\xi)}{2} \right| (x-1)^2 dx$$

$$(2) \text{ 如果对于任何 } x \in [0, 2] \text{ 都有 } |f(x)| \leq 1, |f'(x)| \leq 1$$

$$\text{证明: 对于任何 } x \in [0, 2] \text{ 都有 } |f(x)| \leq 2.$$

$$f(0) = f(x) + f'(x)(0-x) + \frac{f''(\xi)}{2}(0-x)^2 \quad \text{①, } \xi \in (0, x)$$

$$f(2) = f(x) + f'(x)(2-x) + \frac{f''(\xi)}{2}(2-x)^2 \quad \text{②, } \xi \in (x, 2)$$

$$\text{②} - \text{①: } f(2) - f(0) = 2f'(x) + \frac{f''(\xi)}{2}(2-x)^2 - \frac{f''(\xi)}{2}(0-x)^2$$

$$2f'(x) = f(2) - f(0) + \frac{f''(\xi)}{2}(x-x)^2 - \frac{f''(\xi)}{2}(2-x)^2$$

$$2|f'(x)| \leq |f(2)| + |f(0)| + \left| \frac{f''(\xi)}{2} x^2 - \frac{f''(\xi)}{2} (2-x)^2 \right|$$

$$\leq |f(2)| + |f(0)| + \frac{1}{2} |f''(\xi)| x^2 + \frac{1}{2} |f''(\xi)| (2-x)^2$$

$$\leq 2 + \frac{1}{2} [(2-x)^2 + x^2]$$

$$\leq 2 + 2 = 4.$$

$$2|f'(x)| \leq 4, |f'(x)| \leq 2.$$

拓展: $f(x)$ 在 $(0, +\infty)$ 有二阶导数, 如果对于 $x \in (0, +\infty)$

$$\text{都有 } |f(x)| \leq A, |f'(x)| \leq B, \text{ 其中 } A, B \text{ 为正常数, 证明: } |f'(x)| \leq 2\sqrt{AB}$$

$$\text{证明: } |f'(x)| \leq 2\sqrt{AB}$$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(\xi)}{2}(x-x_0)^2, \text{ 令 } h = x-x_0$$

$$f(x_0+h) = f(x_0) + f'(x_0)h + \frac{f''(\xi)}{2}h^2 \quad \text{①}$$

$$f(x_0-h) = f(x_0) - f'(x_0)h + \frac{f''(\xi)}{2}h^2 \quad \text{②}$$

$$\text{①} - \text{②: } f(x_0+h) - f(x_0-h) = 2f'(x_0)h + \frac{1}{2}[f''(\xi_1) - f''(\xi_2)]h^2$$

$$2f'(x_0)h = f(x_0+h) - f(x_0-h) - \frac{1}{2}[f''(\xi_1) - f''(\xi_2)]h^2$$

$$2|f'(x_0)h| \leq |f(x_0+h)| + |f(x_0-h)| + \frac{1}{2}|f''(\xi_1)|h^2 + \frac{1}{2}|f''(\xi_2)|h^2$$

$$\leq 2A + \frac{1}{2}Bh^2. \text{ 不妨设 } h > 0 \text{ (} h < 0 \text{ 同理)}$$

$$Bh^2 + 2A - 2|f'(x_0)|h \geq 0, B \neq 0, \Delta = 4|f'(x_0)|^2 - 8AB \leq 0$$

$$|f'(x_0)| \leq \sqrt{2AB}$$

$$f(x_0+h) = f(x_0) + f'(x_0)h + \frac{f''(\xi)}{2}h^2, \xi \in (x_0, x_0+h)$$

$$f'(x_0)h = f(x_0+h) - f(x_0) - \frac{f''(\xi)}{2}h^2$$

$$|f'(x_0)h| \leq |f(x_0+h)| + |f(x_0)| + \frac{|f''(\xi)|}{2}h^2$$

$$\leq 2A + \frac{1}{2}Bh^2, \frac{1}{2}Bh^2 - |f'(x_0)h| + 2A \geq 0$$

$$\Delta = (f'(x_0))^2 - 4AB < 0, f'(x_0) < 2\sqrt{AB}, \text{即 } f'(x) < 2\sqrt{AB}$$

识别标志: 平方可尝试配方, 多次导一定泰勒

一般为2次泰勒后相减消去某相同项, x 所连续导

最大值: 待给中值定理, 设最大值为 M , 将含 x 的

替换为 M

eg10. 设 $f(x)$ 在 $[0, 1]$ 上连续, 在 $(0, 1)$ 内可微, 并

且 $f(1) = k \int_0^1 x e^{-x} f(x) dx (k > 1)$, 证明至少存在一点

$\xi \in (0, 1)$ 使 $f(\xi) = (1 - \frac{1}{k})f(1)$

令 $g(x) = x e^{-x} f(x)$, $g'(x) = e^{-x} f(x) - x e^{-x} f'(x) + f(x)$

$x e^{-x} = (1-x)e^{-x} f(x) + f(x)$, $x e^{-x} = e^{-x} (1-x)f(x)$

$+ x f'(x)$, $g(1) = f(1)$, $f(1) = k \int_0^1 g(x) dx$, $\exists \eta \in (0, 1)$

$f(1) = k g(\eta) \int_0^1 dx = k g(\eta) \cdot \frac{1}{k} = g(\eta)$, 故 $f(1)$

$= g(\eta) = g(1)$, $\exists \xi \in (\eta, 1)$, 且 $\xi \in (0, 1)$, 使得

$g'(\xi)(1-\eta) = 0$, $g'(\xi) = 0$, $(1-\xi)f(\xi) + \xi f'(\xi) = 0$

$\xi f'(\xi) = (\xi-1)f(\xi)$, 左右同除 ξ 有:

$f'(\xi) = (1-\frac{1}{\xi})f(\xi)$

eg11. 求极限 $\lim_{n \rightarrow \infty} \frac{1}{n} \int_0^n |\sin((t-t_0)\pi)| dt$, 这里

$[x]$ 为不超过 x 最大整数

$\int_n^{n+1} |\sin((t-t_0)\pi)| dt = \int_0^1 |\sin((t-t_0)\pi)| dt$

令 $t-t_0 = u$, 原式 $= \int_0^1 |\sin t \pi| dt = \int_0^1 \sin(\pi t) dt$

$= \frac{1}{\pi} \int_0^\pi \sin u du = \frac{1}{\pi} (-\cos u) \Big|_0^\pi = \frac{2}{\pi}$

由于被积函数 ≤ 1 , $\frac{1}{n} \int_0^n |\sin((t-t_0)\pi)| dt \leq$

$\frac{1}{n} \int_0^n 1 dt = \frac{n}{n} = 1$

而原式 $\leq \frac{1}{n} \int_0^n |\sin((t-t_0)\pi)| dt = \frac{n}{n} = 1$

4

由夹逼定理, $\lim_{n \rightarrow \infty} \frac{1}{n} \leq \text{原式} \leq \lim_{n \rightarrow \infty} \frac{n+1}{n} = 1$

即原极限 $= \frac{2}{\pi}$

eg12. $f(x)$ 在 $\mathbb{R}$ 上 2π -周期连续可导函数, 最小

2π 周期为 2π , 且 $|f'(x)| \leq 1$, 证明: 对任意连续函

数 $g(x)$ 有 $|\int_0^{2\pi} g(x) f'(2x) dx| \leq \max_{x \in [0, 2\pi]} g(x) - \min_{x \in [0, 2\pi]} g(x)$

易知 $\max_{x \in [0, 2\pi]} g(x) \geq \min_{x \in [0, 2\pi]} g(x)$, 且 $\max_{x \in [0, 2\pi]} g(x) - \min_{x \in [0, 2\pi]} g(x) \geq 0$

$g(x) \geq 0$, 只要证 $|\int_0^{2\pi} g(x) f'(2x) dx| \leq 0$, 而绝对值

大于等于 0, 故 $\int_0^{2\pi} g(x) f'(2x) dx = 0$

$\int_0^{2\pi} g(x) f'(2x) dx = g(\xi) \int_0^{2\pi} f'(2x) dx, \xi \in (0, 1)$

$= g(\xi) \int_0^{2\pi} g f'(u) du \cdot \frac{1}{2\pi} = \frac{g(\xi)}{2\pi} \cdot f(u) \Big|_0^{2\pi}$

而 $f(x)$ 周期为 2π , 故 $f(u) \Big|_0^{2\pi} = 0$, 原式 $= 0$

以上各式成立, 证毕

技巧: $\int_a^b f(x) dx = f(x) \Big|_a^b$ 同周期函数

eg13. 证明: $\int_0^\pi x f(\sin x) dx = \pi \int_0^\pi f(\sin x) dx$

① $\int_0^\pi \frac{x \sin x}{1 + \cos x} dx$ 换元: x 或 $\frac{\pi}{2}$

证:

① 令 $t = \pi - x$, $x = \pi - t$, $\int_\pi^0 (\pi - t) f(\sin t) dt$

$= \int_0^\pi (\pi - t) f(\sin t) dt = \int_0^\pi \pi f(\sin t) dt - \int_0^\pi t f(\sin t) dt$

dt , 令 $g(t) = \int_0^\pi t f(\sin t) dt$, 令 $u = \frac{\pi}{2} - t$, $t = \frac{\pi}{2} - u$

$-\int_{\frac{\pi}{2}}^{\frac{\pi}{2}} (\frac{\pi}{2} - u) f(\cos u) du = \frac{\pi}{2} \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} f(\cos u) du$

$= \pi \int_0^\pi f(\cos u) du = \pi \int_0^\pi f(\sin u) du$

$\int_0^\pi \pi f(\sin t) dt = \pi \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} f(\cos u) dt = \pi \int_{\frac{\pi}{2}}^{\frac{\pi}{2}} f(\cos u) du$

$= 2\pi \int_0^\pi f(\cos u) du = 2\pi \int_0^\pi f(\sin u) du$

原式 $= 2\pi \int_0^\pi f(\sin u) du - \pi \int_0^\pi f(\sin u) du$

$= \pi \int_0^\pi f(\sin u) du = \pi \int_0^\pi f(\sin x) dx$, 证毕

② 令 $f(x) \sin x = \frac{\sin x}{1 + \cos x} = \frac{\sin x}{2 - \sin^2 x}$, 令 $\sin x = y$

由①, 原式 $= \pi \int_0^\pi \frac{\sin x}{2 - \sin^2 x} dx = \pi \int_0^\pi \frac{d(\cos x)}{2 - \sin^2 x}$

$= -\pi \int_0^\pi \frac{d(\cos x)}{\cos^2 x + 1} = -\pi \arctan t \Big|_0^\pi = -\pi \arctan 1 - (-\pi \arctan 0)$

$= -\frac{\pi}{4}$



eg 14. 设 $f(x)$ 在闭区间 $[a, b]$ 上具有连续的
 导数且 $f(a) > 0, f'(x) > 0$, 求证存在唯一的 $\xi \in (a, b)$ 使得由 $y=f(x), x=b, y=f(\xi)$ 围成
 的面积 S_1 与 $y=f(x), x=a, y=f(\xi)$ 围成
 的面积 S_2 之比为 2024.

已知 $f(x)$ 单调递增, 故令其原函数为 $F(x)$
 而 $f(a) > 0, f'(x) > 0$, 易知 $F(x)$ 在 $[a, b]$ 上

$$S_1 = \int_{\xi}^b [f(a) - f(\xi)] dx, S_2 = \int_a^{\xi} [f(\xi) - f(x)] dx$$

令 $g(t) = 2024 S_2 - S_1$, 只需证 $g(x)$ 存在唯一零点

$$g(t) = 2024 \int_a^t [f(t) - f(x)] dx - \int_t^b [f(x) - f(t)] dx$$

$$= 2024 [f(t)(t-a) - F(t) + F(a)] - [F(b) - F(t) - f(t)(b-t)]$$

$$F(t) - f(t)(b-t) = 2024 f(t)(t-a) - 2024 F(t) + 2024 F(a) - F(b) + F(t) + f(t)(b-t)$$

$$(b-t) = f(t)[2024(t-a) + (b-t)] - 2023 F(t) + 2024 F(a) - F(b)$$

$$g(a) = (b-a)f(a) - 2023 F(a) + 2024 F(a) - F(b)$$

$$= (b-a)f(a) + F(a) - F(b) = \int_a^b [f(a) - f(x)] dx < 0, F(x) = 2 \int_0^x f(t) dt, f(x) = 2x f^2(x)$$

$$g(b) = 2024 \int_a^b [f(b) - f(x)] dx > 0$$

$g(a), g(b) < 0$, 故 $g(t)$ 在 $[a, b]$ 上至少存在一零点

下通过单调性证明其唯一:

$$g'(t) = 2024 [f'(t) + f(t) - f(t)] - [f(t) + f'(t) + f(t)] = 2023 f(t) > 0, \text{故单调}$$

以上各式成立, 证毕

eg 15. 单调有界: 设 $f(x)$ 在 $(0, +\infty)$ 上二阶导数

连续, 且 $\forall x \in (0, +\infty)$ 都有 $f(x) > 0$ 以及 $f'(x) \leq 0$

又存在正数 M 使得 $|f''(x)| \leq M, (x \in (0, +\infty))$,

已知 $\lim_{x \rightarrow +\infty} f'(x)$ 存在, 证明 $\lim_{x \rightarrow +\infty} f(x) = 0$.

证: $f'(x) \leq 0$ 说明 $f(x)$ 单调递减, 而 $f(x) > 0$
 递减

证明 $f(x)$ 单调递减有下界

$$\lim_{x \rightarrow +\infty} f(x+1) - f(x) = 0$$

$$f'(x) = \lim_{x \rightarrow +\infty} \frac{f(x+1) - f(x)}{(x+1) - x} = 0, \text{证毕}$$

$\lim_{x \rightarrow +\infty}$ 没有用 $|f''(x)| \leq M$? 若没有用单调有

界而是运用泰勒公式?

$$\text{设 } x - x_0 = h, x = x_0 + h, h \rightarrow 0^+$$

$$f(x+h) = f(x) + f'(x)h + \frac{f''(\xi)}{2}h^2$$

$$f'(x)h = f(x+h) - f(x) - \frac{1}{2}f''(\xi)h^2$$

$$f'(x) = \frac{f(x+h) - f(x)}{h} - \frac{1}{2}f''(\xi)h$$

$$\lim_{x \rightarrow +\infty} f'(x) = -\frac{1}{2}f''(\xi)h$$

$$-\frac{M}{2}h \leq -\frac{1}{2}f''(\xi)h \leq \frac{M}{2}h$$

$$\text{当 } h \rightarrow 0 \text{ 时}, f''(\xi)h = 0, Mh \leq f''(\xi)h \leq Mh$$

$$-\frac{1}{2}f''(\xi)h \lim_{x \rightarrow +\infty} f(x) = 0, \text{证毕}$$

eg 16. 设 $f(x)$ 在 $\mathbb{R}$ 上可导且 $f(0) = 0, f'(x) > 0$, 证明:

$$(\int_0^x f(t) dt)^2 \leq 2 \int_0^x t f^2(t) dt$$

$$\text{令 } F(x) = (\int_0^x f(t) dt)^2 = 2 \int_0^x t f^2(t) dt, F(0) = 0$$

$$\exists \xi \in (0, x), \text{使 } x f(\xi) = \int_0^x f(t) dt, \text{则} F(x) =$$

$$F'(x) = 2x f(x) f(\xi) - 2x f^2(x)$$

$$= 2x f(x) [f(\xi) - f(x)], f(x) \text{ 单调递增, 故 } f(\xi) \leq f(x)$$

$$\text{当 } x > 0 \text{ 时}, \xi \in (0, x), f(\xi) \leq f(x), \text{原式} \leq 0$$

$$\text{当 } x < 0 \text{ 时}, \xi \in (x, 0), f(\xi) \geq f(x), \text{原式} \leq 0$$

$$\therefore F'(x) \leq 0, F(x) \text{ 单调递减}$$

$$\therefore F(x) \text{ 在 } (-\infty, 0) \text{ 上 } > 0, (0, +\infty) \text{ 上 } \leq 0$$

$$F(x) \text{ 在 } (-\infty, 0) \uparrow, (0, +\infty) \downarrow, F(x) \leq F(0) = 0$$

原命题得证

若是题目中没有说定义域为 $(0, +\infty)$, 则 $f(x)$

大于 0 或小于 0 的情况都要考虑



求极限

$$\lim_{x \rightarrow 0} \left(\frac{\arcsin x}{\arctan x} \right)^{\frac{1}{\cos x - 1}}$$

$$\text{原式} = \lim_{x \rightarrow 0} \exp \left[\frac{1}{\cos x - 1} \ln \left(\frac{\arcsin x}{\arctan x} \right) \right]$$

$$= \lim_{x \rightarrow 0} \exp \left[\frac{\arcsin x - \arctan x}{\cos x - 1} \right]$$

$$= \exp \lim_{x \rightarrow 0} \frac{\arcsin x - \arctan x}{\cos x - 1}$$

$$= \exp \lim_{x \rightarrow 0} \frac{\frac{1}{\sqrt{1-x^2}} - \frac{1}{1+x^2}}{-2x}$$

$$= \exp \lim_{x \rightarrow 0} \frac{\frac{1-x^2}{\sqrt{1-x^2}} - \frac{1}{1+x^2}}{-2x}$$

$$\text{令 } \sqrt{1-x^2} = t, \lim_{x \rightarrow 0} \frac{1-x^2}{\sqrt{1-x^2}} = \frac{1-t^2}{t}, 1-x^2=t^2, x^2=1-t^2$$

$$= -\frac{2}{3} \lim_{t \rightarrow 1} \frac{\frac{1}{t} - \frac{1}{1-t^2}}{1-t^2} = -\frac{2}{3} \lim_{t \rightarrow 1} \frac{1-t^2-t}{(1-t^2)t(1-t^2)}$$

$$= \frac{2}{3} \lim_{t \rightarrow 1} \frac{t^2+t-2}{t(t^2-2)(t-1)(t+1)} = \frac{2}{3} \lim_{t \rightarrow 1} \frac{(t+2)(t-1)}{t(t^2-2)(t+1)}$$

$$= \frac{2}{3} \lim_{t \rightarrow 1} \frac{t+2}{t(t^2-2)(t+1)} = \frac{2}{3} \times \left(-\frac{3}{2}\right) = -1$$

$$\text{原式} = \exp(-1) = \frac{1}{e}$$

$$\text{or: 原式} = \exp \lim_{x \rightarrow 0} \frac{1}{3} \left(1 + \frac{1-\sqrt{1-x^2}}{x^2} \right)$$

eg2. 求极限 $\lim_{n \rightarrow \infty} \frac{n!}{n^n}$

$$\text{原式} = \exp \lim_{n \rightarrow \infty} \left(\frac{1}{n} \ln(n!) - \ln n \right) = \exp \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \ln \left(\frac{i}{n} \right)$$

将 $[0, 1]$ 区间分割为 n 个小区间, 每个区间取长度 $\Delta x = \frac{1}{n}$, 记右端点为每个区间的 ξ_i .

$$\text{原式} = \exp \int_0^1 \ln x dx = \exp (x \ln x |_0^1 - x |_0^1) = \frac{1}{e}$$

eg3. $\lim_{x \rightarrow \infty} (x^3 \ln \frac{x+1}{x-1} - 2x^2)$

$$= \lim_{x \rightarrow \infty} (x^3 \ln (1 + \frac{2}{x-1}) - 2x^2)$$

$$= \lim_{x \rightarrow \infty} (x^3 \left(\frac{2}{x-1} - \frac{1}{2} \left(\frac{2}{x-1} \right)^2 + \frac{1}{3} \left(\frac{2}{x-1} \right)^3 \right) - 2x^2)$$

$$= \lim_{x \rightarrow \infty} \frac{2x^3(x-1)^2 - 2x^3(x-1) + \frac{2}{3}x^3 - 2x^2(x-1)^3}{(x-1)^3}$$

$$= \lim_{x \rightarrow \infty} \frac{\frac{2}{3}x^3 - 2x^2}{(x-1)^3} = \frac{2}{3}$$

eg4. $I_1 = \int \sqrt{1+3\cos^2 x} \cdot \sin(2x) dx = -\int \sqrt{1+3\cos^2 x} \cdot 2\cos x$

$$d(\cos x) = -2 \int \cos x \sqrt{1+3\cos^2 x} d(\cos x), \text{令 } t = \sqrt{3}\cos x$$

$$I_1 = -2 \int \frac{t}{\sqrt{3}} \sqrt{1+t^2} d\left(\frac{t}{\sqrt{3}}\right) = -\frac{2}{3} \int t \sqrt{1+t^2} dt, \text{令 } t = \tan u$$

$$I_1 = -\frac{2}{3} \int \tan u \sqrt{1+\tan^2 u} d(\tan u)$$

$$= -\frac{2}{3} \int \frac{\sin u}{\cos u} \cdot \frac{1}{\cos u} \cdot \frac{1}{\cos u} du = \frac{2}{3} \int \frac{d(\cos u)}{\cos^3 u}$$

$$= \frac{1}{9} \frac{1}{\cos^2 u} = \frac{1}{9} \frac{1}{\cos^2 u} = \frac{1}{9} \frac{1}{\cos^2 u} = \frac{1}{9} \frac{1}{\cos^2 u}$$

$$\cos u = \frac{1}{\sqrt{1+3\cos^2 x}}, I_1 = \frac{1}{9} (1+3\cos^2 x)^{\frac{3}{2}} + C$$

$$I_1 = -\frac{2}{3} \int \frac{\sin u}{\cos u} \cdot \frac{1}{\cos u} \cdot \frac{1}{\cos u} du = \frac{2}{3} \int \frac{d(\cos u)}{\cos^3 u}$$

$$= -\frac{2}{9} \cdot \frac{1}{\cos^2 u} = -\frac{2}{9} (1+3\cos^2 x)^{\frac{3}{2}}$$

eg5. $I_1 = \int \frac{x^2 dx}{(x \sin x + \cos x)^2} = \int \frac{x}{\cos x} \cdot \frac{x \cos x dx}{(x \sin x + \cos x)^2}$

$$= -\int \frac{x}{\cos x} d\left(\frac{1}{x \sin x + \cos x}\right) = -\frac{x}{\cos x (x \sin x + \cos x)}$$

$$+ \int \frac{1}{x \sin x + \cos x} \cdot \frac{\cos x + x \sin x}{\cos^2 x} dx = -\frac{x}{\cos x (x \sin x + \cos x)}$$

$$\cos x + \int \frac{dx}{\cos^2 x} = -\frac{x}{\cos x (x \sin x + \cos x)} + \tan x + C$$

eg6. $I = \int_0^{\frac{\pi}{2}} e^x \frac{1+\sin x}{1+\cos x} dx$

分部积分法 + 三角公式



eg1. $I = \int_0^{+\infty} \frac{1-x^2}{x^4+1} dx$

奇点为 $+\infty$

$$I = \lim_{x \rightarrow +\infty} \int_0^x \frac{1-t^2}{t^4+1} dt$$

$$I = \int \frac{1-t^2}{t^4+1} dt = \int \frac{1}{2} \frac{[(t^2+\sqrt{2}t+1) + (t^2-\sqrt{2}t+1)]}{(t^2+\sqrt{2}t+1)(t^2-\sqrt{2}t+1)} dt$$

$$= 2 \int \frac{dt}{t^4+1} - \frac{1}{2} \int \frac{d(t-\frac{\sqrt{2}}{2})}{(t-\frac{\sqrt{2}}{2})^2+(\frac{\sqrt{2}}{2})^2} - \frac{1}{2} \int \frac{d(t+\frac{\sqrt{2}}{2})}{(t+\frac{\sqrt{2}}{2})^2+(\frac{\sqrt{2}}{2})^2}$$

$$t = \frac{1}{u}, \quad \int \frac{dt}{t^4+1} = \int \frac{d(\frac{1}{u})}{(\frac{1}{u})^4+1} = \int \frac{u^4}{u^4+1} \cdot (-\frac{1}{u^2}) du$$

$$= - \int \frac{u^2}{u^4+1} du$$

